

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

PHY 219 — Past Questions & Materials
by Renance

Course: PHY 219
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 7
Levels: 200

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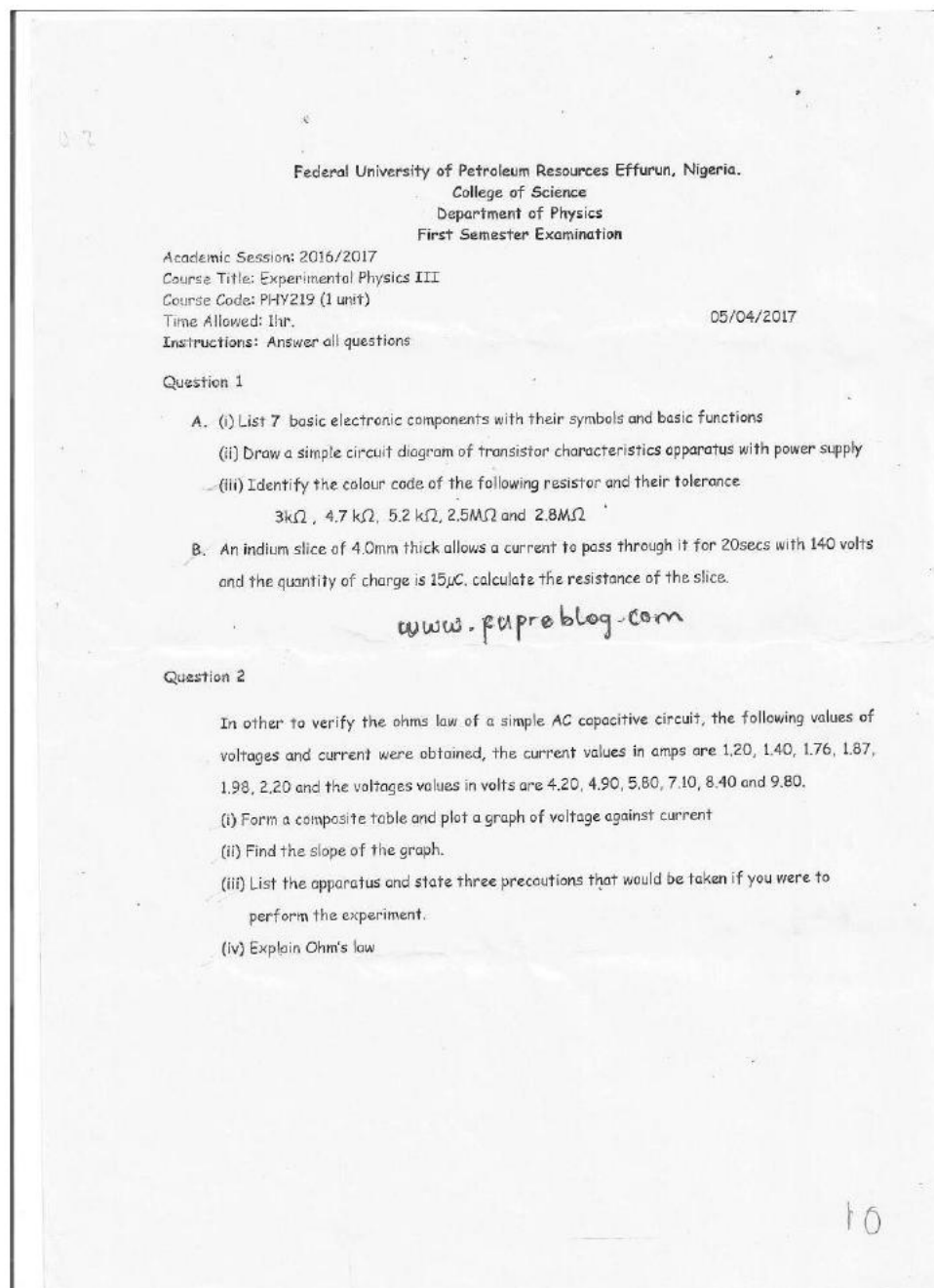
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Paper 1 of 7 — PHY 219 - 2016/2017 - First Semester - Experimental Physics III

Level 200 • First Semester • PastQuestion

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Paper 2 of 7 — PHY 219 - 2016/2017 - First Semester - Experimental Physics III

Level 200 • First Semester • PastQuestion

Federal University of Petroleum Resources Effurun, Nigeria.

College of Science

Department of Physics

First Semester Examination

Academic Session: 2016/2017

Course Title: Experimental Physics IIT

Course Code: PHY219 (1 unit)

Time Allowed: thr. 05/04/2017

Instructions: Answer all questions

Question 1

- A. (i) List 7 basic electronic components with their symbols and basic functions
(ii) Draw a simple circuit diagram of transistor characteristics apparatus with power supply
(ii) Identify the colour code of the following resistor and their tolerance

3k, 4.7 kA, 52k, 25MM and 2.8M2

- B. An indium slice of 4.0mm thick allows a current to pass through it for 20secs with 140 volts and the quantity of charge is 15, calculate the resistance of the slice.

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Question 2

In other to verify the ohms law of a simple AC capacitive circuit, the following values of voltages and current were obtained, the current values in amps are 1.20, 1.40, 1.76, 1.87, 1.98, 2.20 and the voltages values in volts are 4.20, 4.90, 5.80, 7.10, 8.40 and 9.80.

- (i) Form a composite table and plot a graph of voltage against current
(ii) Find the slope of the graph.
(iii) List the apparatus and state three precautions that would be taken if you were to perform the experiment.
(iv) Explain Ohm's law

Paper 3 of 7 — PHY 219 - 2014/2015 - First Semester - EXPERIMENTAL PHYSICS III

Level 200 • First Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES EFFURUN-NIGERIA

COLLEGE OF SCIENCE

DEPARTMENT OF PHYSICS

FIRST SEMESTER EXAMINATION 2014/2015 ACADEMIC SESSION

COURSE TITLE: PHY 219: (EXPERIMENTAL PHYSICS III)

INSTRUCTIONS: ANSWER ALL QUESTIONS TIME: 1 Hr

IFR1 has the colour code of blue, black and gold, — R2 has the colour code of blue, black and gold

R3 has the colour code of yellow, black and gold,

From the circuit diagram above Calculate (i) I_q (Thevenin Current)

(ii) V_{ix} (Thevenin Voltage)

b Calculate the resistor value of a 5 colour band resistor which has the colours as yellow, blye, orange, red and gold respectively.

c Explain how a faulty diode be detected in circuit,

2a, With the aid of diagram explain how an NPN transistor can be tested.

b From a transistor characteristic experiment, explain and also show the graphical representation of the input and output characteristics,

c State two uses of a transistor,

3a

In oscilloscope experiment, suppose a “vertical attenuator” of an AC signal is set to 1.5v/Div and the probe factor is 10, ifthe signal occupies 6 divisions vertically what is the peak—to—peak voltage and its root mean square (Vine)

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R4 has the colour code of brown, black and violet.

b ifthe Horizontal time control of an AC signal Is Set19 9-Ams/Div and a complete cycle of the sine ‘wave occupies 7 divisions: “what isthe signal period (8) and its frequency?

¢ What isa spectrometer and explain its three essential parts

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Paper 4 of 7 — PHY 219 - 2013/2014 - First Semester - EXPERIMENTAL PHYSICS III

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES EFFURUN-NIGERIA

COLLEGE OF SCIENCE

DEPARTMENT OF PHYSICS

FIRST SEMESTER EXAMINATION 2013/2014 ACADEMIC SESSION

COURSE TITLE: PHY 219: (EXPERIMENTAL PHYSICS III)

INSTRUCTIONS: ANSWER ALL QUESTIONS TIME ALLOWED: 1 Hr

A. In order to verify the Ohm's Law of a simple AC capacitive circuit. The following values of voltage and current were obtained, the voltage values in . ~ volts are 2.20, 2.80, 3.00, 3.40, 4.00, 5.00, 6.00 and the current values in amps are 0.25, 0.28, 0.30, 0.35, 0.40, 0.50, 0.60

Questions:

- i. Form a composite table and plot a graph of voltage against current.
- ii, Find the slope of the graph and the value of the Resistor used in the circuit
- iii, If you are to perform the experiment, list the apparatus and 3 precautions that would be taken
- iv. An Indium slice of 2.5mm thick allows a current to pass through it for 20secs with 10volts and the quantity of charge is 1000uc. Calculate the resistance of the slice.

Vv. Explain Ohm's law. a Sa a

B. (i) List 5 basic electronic components with their symbols and basic functions.

(ii) What is the importance of the circuit built by Relay coil and thermal 555

(iii) List three advantages and two disadvantages of Relay coil and thermal circuit over any other circuits.

(iv) Identify the colour code of the given resistors and calculate their resistance values.

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Paper 5 of 7 — PHY 219 2016 2017 EXAM

Level 200 • First Semester • PastQuestion

Federal University of Petroleum Resources Effurun, Nigeria.

College of Science

Department of Physics

First Semester Examination

Academic Session: 2016/2017

Course Title: Experimental Physics TIT

Course Code: PHY219 (1 unit)

Time Allowed: thr. 05/04/2017

Instructions: Answer all questions

Question 1

A. (i) List 7 basic electronic components with their symbols and basic functions

(ii) Draw a simple circuit diagram of transistor characteristics apparatus with power supply

(iii) Identify the colour code of the following resistor and their tolerance

3kΩ, 4.7kΩ, 5.2kΩ, 2.5MΩ and 2.8MΩ

B. An indium slice of 4.0mm thick allows a current to pass through it for 20secs with 140 volts and the quantity of charge is 15C. calculate the resistance of the slice.

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Question 2

In order to verify the ohms law of a simple AC capacitive circuit, the following values of voltages and current were obtained, the current values in amps are 1.20, 1.40, 1.76, 1.87, 1.98, 2.20 and the voltages values in volts are 4.20, 4.90, 5.80, 7.10, 8.40 and 9.80.

(i) Form a composite table and plot a graph of voltage against current

(ii) Find the slope of the graph.

(iii) List the apparatus and state three precautions that would be taken if you were to perform the experiment.

(iv) Explain Ohm's law

Paper 6 of 7 — PHY 219 2014 2015 EXAM

Level 200 • First Semester • PastQuestion

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COLLEGE OF SCIENCE

DEPARTMENT OF PHYSICS

FIRST SEMESTER EXAMINATION 2014/2015 ACADEMIC SESSION

COURSE TITLE: PHY 219: (EXPERIMENTAL PHYSICS III) az

INSTRUCTIONS: ANSWER ALL QUESTIONS - y

R1 R2

TIME: 1 Hr

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If R1 has the colour code of blue, black and gold,

R2 has the colour code of blue, black and gold

R3 has the colour code of yellow, black and gold, R4 has the colour code of brown, black and violet,

From the circuit diagram above Calculate (i) Ity (Thevenin Current)

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Paper 7 of 7 — PHY 219 2013 2014 EXAM

Level 200 • First Semester • PastQuestion

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COLLEGE OF SCIENCE

DEPARTMENT OF PHYSICS

FIRST SEMESTER EXAMINATION 2013/2014 ACADEMIC EorSsiOn

COURSE TITLE: PHY 219: (EXPERIMENTAL PHYSICS III)

INSTRUCTIONS: ANSWER ALL QUESTIONS TIME ALLOWED: 1 Hr

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Questions:

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ii. Find the slope of the graph and the value of the Resistor used in the circuit

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